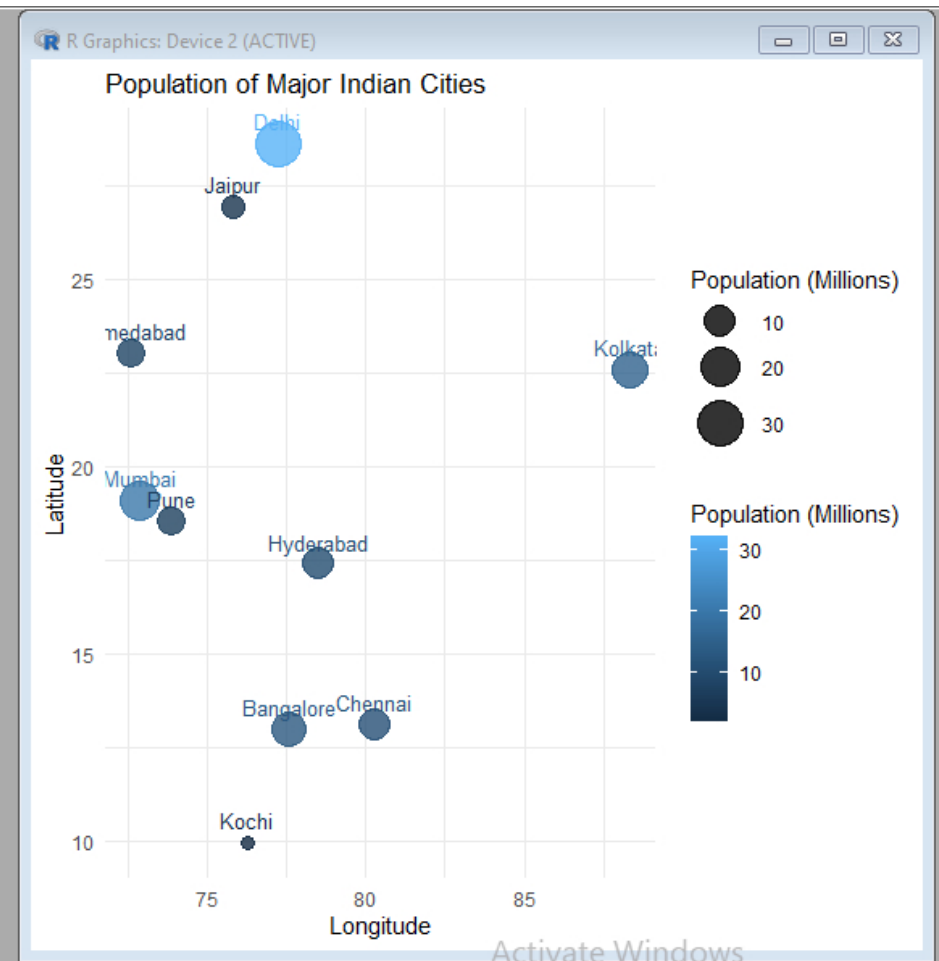


```
R Console
>
> population <- c(11,13,10,20,32,7,15,8,4,2)
>
> geo <- data.frame(city, latitude, longitude, population)
>
> ggplot(
+   18.52,22.57,23.02,26.91,9.93)
+
+
+   longitude <- c(80.27,77.59,78.48,72.87,77.21,
+   73.85,88.36,72.57,75.79,76.26)
+
+   ge
+   ge
+   so
+   la
+
+   population <- c(11,13,10,20,32,7,15,8,4,2)
+
+   geo <- data.frame(city, latitude, longitude, population)
+
+   ggplot(geo, aes(x = longitude,
+   y = latitude,
+   size = population,
+   colour = population)) +
+   geom_point(alpha = 0.8) +
+   geom_text(aes(label = city), vjust = -0.8, size = 3.5) +
+   scale_size(range = c(3,10)) +
+   labs(
+     title = "Population of Major Indian Cities",
+     x = "Longitude",
+     y = "Latitude",
+     colour = "Population (Millions)",
+     size = "Population (Millions)"
+   ) +
+   theme_minimal()
+
+ geo[which.max(geo$population), ]
```



R Console

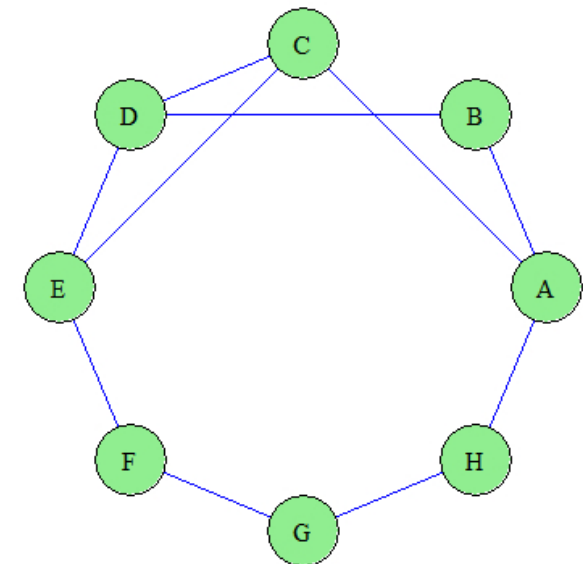
```
> plot(g,  
+   vertex.color = "skyblue",  
+   vertex.size = 30,  
+   vertex.label.color = "black",  
+ )
```

Untitled - R Editor

```
edges <- data.frame(from, to)  
  
> deg  
> print(deg)  
A B C  
3 2 3  
Node v  
  
> plot(g,  
+   vertex.color = "skyblue",  
+   vertex.size = 30,  
+   vertex.label.color = "black",  
+   vertex.label.cex = 1,  
+   edge.color = "gray",  
+   main = "Network Graph")  
  
deg <- degree(g)  
print(deg)  
  
cat("Node with highest degree:", names(which.max(deg)), "\n")  
  
> |  
  
plot(g,  
  layout = layout_in_circle(g),  
  vertex.color = "lightgreen",  
  vertex.size = 30,  
  vertex.label.color = "black",  
  vertex.label.cex = 1,  
  edge.color = "blue",  
  main = "Circular Network Graph")
```

R Graphics: Device 2 (ACTIVE)

Circular Network Graph



Activate Windows


```
R Console
+ x = "Marks",
+ y = "Attendance (%)",
+ size = "Study Hours",
+ colour = "Study Hours"
+ )
+ th
>
> pairs
+
> cor
marks
attend
study_
>
> data
stu
10
>
> heat
+
+
+
> |
<

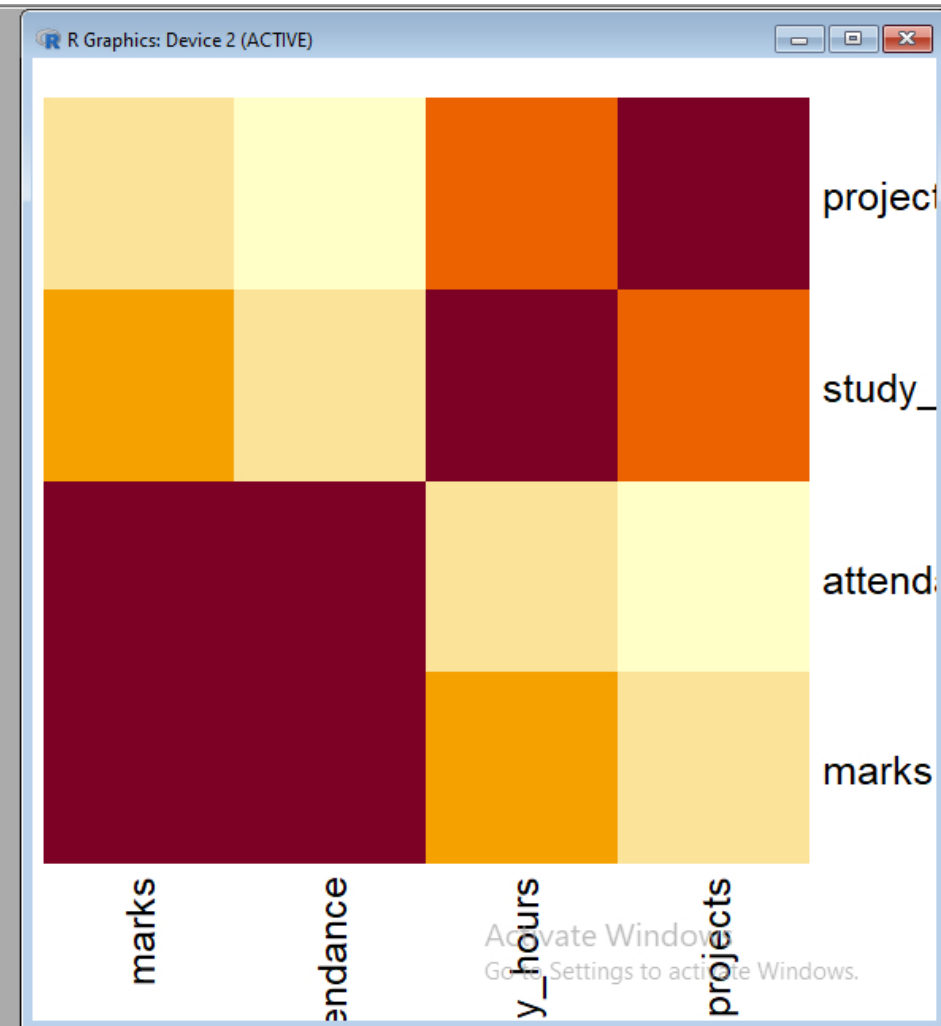
Untitled - R Editor
y = attendance,
size = study_hours,
colour = study_hours)) +
geom_point(alpha = 0.8) +
geom_text(aes(label = student), vjust = -1) +
scale_size(range = c(4,10)) +
labs(
  title = "Student Performance",
  x = "Marks",
  y = "Attendance (%)",
  size = "Study Hours",
  colour = "Study Hours"
) +
theme_minimal()

pairs(data[,c("marks", "attendance", "study_hours", "projects")],
      main = "Scatter Plot Matrix")

cor(data[,c("marks", "attendance", "study_hours")])

data[which.max(data$marks),]

heatmap(cor(data[,c("marks", "attendance", "study_hours", "projects")]),
        Rowv = NA,
        Colv = NA,
        scale = "none")
```



```
R Console
+ x = "Sales",
+ y = "Profit",
+ size = "Customers",
+ colour = "Customers"
+ )
+ )
> cor
> pair

sales
profit
customers
branches

> pair
+
> if(c
+
+ cor(data[, c("sales", "profit", "customers", "branches")])
+ } el
+ }
Incre
> |

Untitled - R Editor
aes(x = sales,
  y = profit,
  size = customers,
  colour = customers)) +
geom_point(alpha = 0.8) +
scale_size(range = c(4,10)) +
labs(
  title = "Sales vs Profit",
  x = "Sales",
  y = "Profit",
  size = "Customers",
  colour = "Customers"
) +
theme_minimal()

cor(data[, c("sales", "profit", "customers", "branches")])

pairs(data[, c("sales", "profit", "customers", "branches")],
  main = "Scatter Plot Matrix")

if(cor(data$branches, data$sales) > 0 &
  cor(data$branches, data$profit) > 0){
  cat("Increase in branches is associated with higher sales and profits.\n")
} else {
  cat("No positive association found.\n")
}
```

